

Ian Leung

☎ (+1) 647-763-3121 | ✉ i24leung@uwaterloo.ca | 🌐 ianleung.com | 🐙 github.com/ianleung12 | 💼 linkedin.com/in/ianleung12

EXPERIENCE

Roblox

Jun 2026 – Aug 2026

Software Engineer Intern (Network Reliability) | Go, Python

San Mateo, CA

- Automated the Network Engineer deployment process, pushing to **5,000+** production devices in global datacenters
- Built a GitHub Actions CI/CD pipeline, saving **400+ engineering hours** annually by eliminating manual tasks
- Orchestrated Temporal Workflows with LLM analysis & slack alerting for safe, autonomous deployment of network changes, preventing outages/misconfigurations and identifying issues on **20+** network devices
- Identified and fixed a resource leak in an internal network service, recovering **7%** CPU utilization across devices

WQC

Jan 2026 – Present

Software Engineer (DevOps) | Rust, Python, Docker

Waterloo, ON

- Launched a Grafana observability platform with **100+** metrics, including membership and infrastructure monitoring
- Harnessed Alertmanager to send automatic Discord alerts for critical service failures, maintaining a **>99.5%** uptime
- Engineered portal.waterlooquantclub.com, a full-stack club recruitment & event portal hosted on AWS for **700+** users
- Built and deployed [2v2bluff](#), a real-time multiplayer **Rust/WebSocket** card game supporting four concurrent players

Sunnybrook Research Institute

Jul 2025 – Aug 2025

Software Research Intern | Python

Toronto, ON

- Built an event tracing, playback, and logging tool that records and simulates clinician workflows in **Focused Ultrasound** software, surfacing **5+** defects and edge cases in real usage patterns
- Automated regression testing by generating synthetic JSON event flows and validating results against recorded baselines, cutting testing time by **91%**

PROJECTS

Pharmagotchi – Pharmacy Companion App | Kotlin, Android Studio, Figma, OpenRouter

- Built a Tamagotchi-inspired Android companion app to track health & medications and connect users with physicians
- Integrated an **OpenRouter**-powered chat assistant with Retrieval-Augmented Generation (RAG) that parses health conditions and medications into structured JSON vitals and provides context-aware medication guidance
- Connected Fitbit tracking devices through the Google Health API, centralizing health data for seamless monitoring

PeerAssist – Peer Review Platform | Java, Supabase, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for **20+** students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with Apache PDFBox and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

N-body Simulation | Python, Pyglet, NumPy

- Built a 2D particle gravity simulation with tunable constants, supporting **1000+** concurrent particles
- Implemented a Barnes-Hut quadtree algorithm, cutting unnecessary calculations and speeding up simulations by **220+%**
- Designed a graphics renderer to display particles and quadtree bounding boxes using pyglet, an OpenGL binding

SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash

Technologies Temporal, gRPC, REST, PyTorch, Pandas, Grafana, PostgreSQL, MongoDB, React, Actions

Tools Linux, Git, Docker, AWS (EC2, S3, Lambda, etc.), OpenRouter, Jupyter, Supabase, Android Studio

EDUCATION

University of Waterloo

GPA: 3.8/4.0

Bachelor of Computer Science, Finance, & Statistics – Triple Major

- President's Scholarship of Distinction (**\$5000**), Undergraduate Achievement Award (**\$1500**)
- Relevant Coursework: Object-Oriented Programming (Advanced), Sequential Programming (Advanced), Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Data Analytics